

Degree Program and Timeline:

- Are you completing the honors thesis in a major or minor?: **major**
- What major/minor are you completing your honors thesis in?: **Data Science**
- Which will be your first semester of honors thesis?: **Spring 2026 Second Seven Week**
- Which will be your second semester of honors thesis?: **Spring 2027**
- When is your intended graduation semester?: **Spring 2027**

Faculty Thesis Mentor info:

- Faculty Thesis Mentor's first and last name: **Joseph Watkins**
- Faculty Thesis Mentor's University of Arizona email: **jwatkins@arizona.edu**
- What is your Thesis Mentor's primary department?: **Math/Data Science**

What is the topic of your honors thesis?

Briefly describe what you plan to study, create, or investigate. Your description should clearly explain:

- The central question, problem, or idea you are exploring
- The general approach or method you expect to use
- How this project fits within your discipline

I plan to study the impacts of two U.S. tariff waves using a comparative analysis to see if the impacts of the 2018-2019 tariffs will be similar to the 2025-2026 tariffs. I will specifically look at how the 2018-2019 U.S. tariff increases affected small businesses versus large corporations by looking at cash flow and closures. I will compare this to what is happening currently to see if businesses are being affected similarly or may be in the future. This falls into the central question: Do periods of tariff volatility and uncertainty affect small businesses with bankruptcies and closures differently than they do larger corporations?

I will answer this question by using data analysis and regression. For the 2018 period, I will use the Census Bureau's Business Dynamics Statistics. For current data, I will look at bankruptcy filings from the American Bankruptcy Institute and tariff tracking from the Tax Policy Center to compare trends. I will compare the survival rates of small vs large businesses in areas like electronics and agriculture that are more heavily tariffed. I will use a linear regression model with a combined data set to see if the amount of tariffs, sector, and size of business are strong predictors for closures in 2018-2019 and compare that to 2025-2026. If patterns are the same between the two periods, the predictors for 2018-2019 may be the same for 2025-2026.

As a data science major, this project combines many things that I have learned in my classes, from data wrangling and cleaning to linear regression and other statistical methods. I also plan to get a Master of Science in Business Analytics and work in data analytics for a business, so this gives me experience in economics and businesses as well. I

currently have an internship with a tax company that deals with a lot of finances, so it fits with that as well.

What are the academic, professional, and personal contexts for this project?

Explain why this project matters to you and your field. You may address:

- How your project connects to existing scholarship, creative work, or practice
- What gap, question, or need your project is responding to
- How this project relates to your academic goals, career interests, or personal motivations

Though the 2018-2019 tariffs have been studied a lot and there is a lot of economic literature on them, there is a gap in how they can correlate to the 2025-2026 tariffs. The extreme 2018-2019 tariffs were largely with China, and the 2025-2026 tariffs were a lot broader. However, because of how many imports we had from China, there could still be a correlation between what happened to businesses then and now. There is also a new factor to the 2025-2026 tariffs that the Supreme Court has found some illegal, and there is a gap in looking at how factors like these will affect business outcomes. Analyzing real-time data will include looking at current datasets instead of just looking at cleaned historical data, which gives valuable hands-on experience.

What form will the final product(s) of your honors thesis take? Be as specific as possible. This should include:

- The type of final product (for example: literature review, creative work, portfolio, performance, exhibition, report, a combination of these, etc.)
- Any expected length, format, or structure discussed with your mentor
- Disciplinary standards that apply to your project (citations, documentation, etc.)

This section should clearly reflect a shared understanding between you and your mentor about what you are working toward. This may change as the project evolves; articulating it now ensures mentor and mentee are on the same page.

The final product would be a report with an abstract, introduction, methods, findings, conclusion, and appendix, and will include the data I used, my process of working with the data, statistical methods I carried out, and any visualizations that help with my conclusion. This will be roughly 8-15 pages. I will cite any of the data sources/sets I used and economic sources about tariffs.

First Semester Work Plan

What is your work plan for the first semester of the honors thesis?

Describe what you expect to complete during your first thesis semester. Consider including:

- Major tasks, milestones, or deliverables
- Research, preparation, or exploratory work
- Approximately how you will pace the work across the semester

For the first semester, I will work on collecting data from the databases I talked about previously. I will ensure that the data is clear and aggregate any different datasets as necessary. I will plan out the statistical and linear regression methods that will work best for the data and begin to compare the current data with data from 2018-2019. For the first week, I will work on finding and downloading data sets. From there, I will combine any data necessary and ensure that the data for 2025-2026 can be compared with previous data by cleaning and changing column names. The later part of the semester will look at initial linear models. Throughout this process, I will also document what I am doing to begin writing the introduction and methods section. This semester will focus on data gathering, initial coding, and analysis.

5. Second Semester Work Plan

What is your work plan for the second semester of the honors thesis?

Describe what you expect to complete during the second semester. Consider including:

- Major writing, creation, analysis, or revision tasks
- Approximately how you will pace the work across the semester
- Completion of the final product
- Plans for submission, presentation, or documentation

The second semester will finalize the regression methods, create visualizations to show conclusions, and add the abstract, findings, conclusions, and next steps to my report. I will also create a presentation with this information in a concise, informative way to show my findings.

How firm or flexible are your project deadlines, as agreed upon with your mentor?

Briefly explain:

- Which deadlines are fixed and which may be adjusted
- How changes to the timeline will be communicated and approved

The deadline for data needs to be found and cleaned by the end of the first semester is fixed, but what initial methods I use can be adjusted. Changes to the timeline can be communicated through email and meetings.

How often will you meet with your faculty thesis mentor, and in what format? Include:

- Expected meeting frequency
- Format of meetings (in person, Zoom, email check-ins, etc.)
- Expectations for communication between meetings

Meetings will occur every other week in person or on Zoom. Between meetings, communication will be by email.

How will your work be assessed and graded across the honors thesis experience?

Be specific about:

- How and when you will receive feedback
 - What criteria your mentor will use to evaluate progress
 - Expectations for drafts, feedback, revisions, or intermediate deliverables
- (Remember: you receive a separate letter grade for each semester of honors thesis.)

I will receive feedback in our meetings and through email when a goal is met. The criteria my mentor will use are typical data report criteria methods to evaluate mine, such as if the introduction and methods explain the goal and process well, if statistical methods are used correctly, and if the findings correctly communicate the intended goal.

How will you plan your time to meet the honors thesis requirement of at least 270 total hours?
Explain:

- Your expected weekly or monthly time commitment
- How you plan to balance thesis work with coursework and other obligations

The total weeks are 22.5, with one semester being only 7.5 weeks, so the weekly time commitment will be 12 hours for each semester. I currently have a 7.5-week course ending, so this semester I will be able to start my thesis work, for which I previously did work in that class. I will balance my time with studying for my classes and also working on my thesis. For the second semester, I will have the same hours commitment and ensure I am doing the same thing with new classes. I will treat it like a class I need to study and do outside work for, and plan my time accordingly.

What resources or approvals will this project require?

This may include:

- Equipment, materials, or specialized software
- Training or skills you will need to acquire
- Use of human subjects and potential IRB approval
- Any other approvals or access needed to complete the project

This will require the software R and any associated packages for R, along with access to any datasets I do not already have, but all should be publicly accessible. I will need to use statistical and coding skills that I already have, and there are no human subjects.

What ethical considerations apply to your project, and how will they be addressed?

Briefly describe:

- Relevant disciplinary standards for citations, sources, or creative work
- Plans for ethical use of data, sources, or human participants
- Conversations you have had with your mentor about these expectations, including clear policies regarding the use of generative AI at any/all stages of the project

Ethical considerations in data science include clear citation of any data that is being used, where it came from, and how it is included. It also includes taking into account any

biases that may be present. AI will not be planned to use at any stage of this project, but if there is any coding or debugging that is used with AI for any reason, it will also be cited. Everything will be cited in the citations area, and any code can be cited in the appendix.